SIMATS ENGINEERING

SWETHA.B

192412215

EXPERIMENT 11

ITA0606

MACHINE LEARNING

Write a program for the task of Credit Score Classification

Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

# Read dataset
df = pd.read_csv("/content/credit_score.csv")

# Remove missing values
df = df.dropna()

# Separate features and target
X = df.iloc[:, :-1]
y = df.iloc[:, -1]

# Convert categorical features into numbers
X = pd.get_dummies(X)

# Encode target classes
encoder = LabelEncoder()
y = encoder.fit_transform(y)

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create and train classifier
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
```

```
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Accuracy (%):", accuracy * 100)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=encoder.classes_
).plot()

plt.title("Credit Score Classification")
plt.show()
```

OUTPUT:

```
Accuracy: 0.3333333333333333
Accuracy (%): 33.33333333333333

Confusion Matrix:
[[0 1]
 [1 1]]
```